

Project Name: Example House

Address: 16 South Main Street, Monroe, OH

OUTDOOR DESIGN CONDITIONS

Weather station: Middletown (Dayton DD)

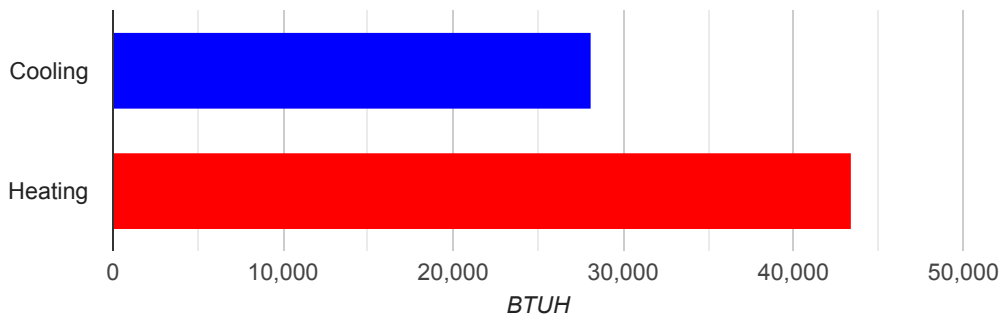
Summer Outdoor F:	90	Summer Indoor F:	75	Design Grains:	26	Daily Range:	Medium
Winter Outdoor F:	5	Winter Indoor F:	70	Cooling RH:	50	Elevation (Ft):	651

LOAD CALCULATION TOTALS

HVAC System: Main

Heated square footage:	1,933	Heating BTUH:	43,426
Cooled square footage:	1,933	Cooling BTUH:	28,028
Heated volume (above grade CF):	17,399	CFM:	1,061
Cooled volume (above grade CF):	17,399	Sensible cooling:	23,352
Exposed wall area (SF):	1,776	Latent cooling:	4,676
		SHR:	0.833

Load Calculation



Approved ACCA MJ8 Calculations

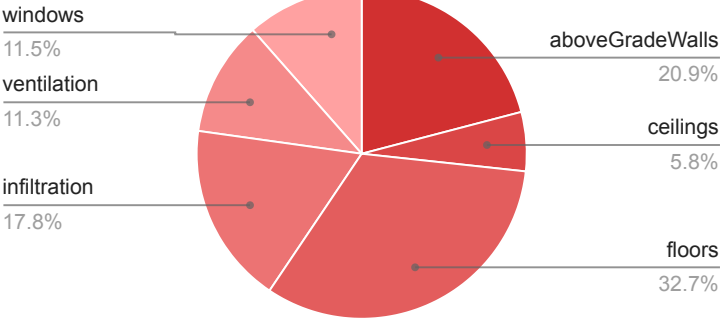
Calculations are based on the ACCA Manual J 8th Edition and are approved by ACCA. All computed calculations are estimates on building use, weather data, and inputted values such as R-Values, window types, duct loss, etc. Equipment selections should meet both the latent and sensible gain as well as building heat loss.



HEATING AND COOLING LOADS

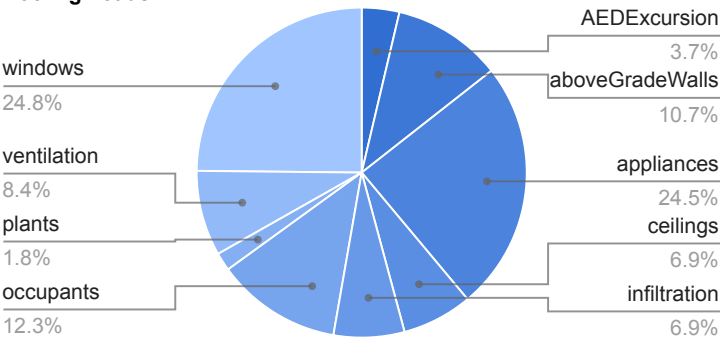
HEATING LOADS		
SECTION	AREA	HEAT LOSS
aboveGradeWalls	1,536	9,085
ceilings	1,933.2	2,513
floors	1,933.2	14,210
infiltration	0	7,721
skylights	0	0
ventilation	0	4,905
windows	240	4,992
Totals		43,426

Heating Loads



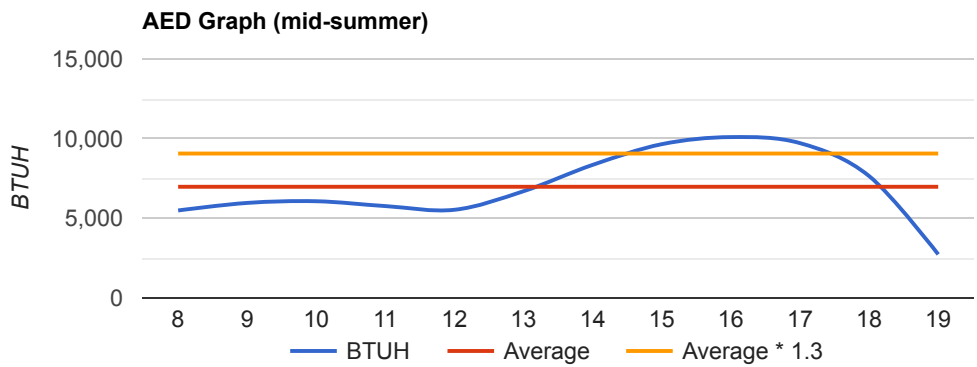
COOLING LOADS			
SECTION	AREA	SENSIBLE	LATENT
AEDExcursion	0	1,033	0
aboveGradeWalls	1,536	3,005	0
appliances	0	6,511	358
ceilings	1,933.2	1,933	0
floors	1,933.2	0	0
infiltration	0	938	1,005
occupants	0	1,840	1,600
plants	0	0	500
skylights	0	0	0
ventilation	0	1,132	1,213
windows	240	6,960	0
Totals		23,352	4,676

Cooling Loads

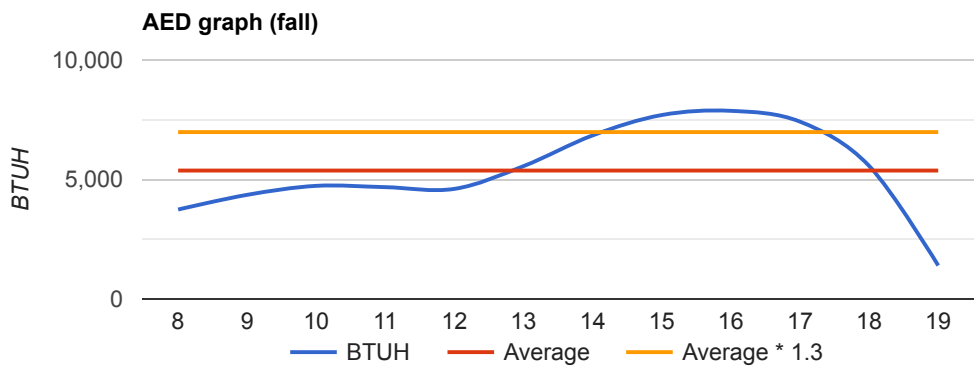


FENESTRATION LOADS

Warning (0): This application has glass areas that produced relatively large cooling loads for part of the day. Zoning may be required to overcome spikes in solar load for one or more rooms. A zoned system may be required, or some rooms may require zone control (provided by individual, motorized, thermostatically controlled dampers).



This graph represents hourly aggregate fenestration loads in mid-summer.



This graph represents hourly aggregate fenestration loads in October.

COMPONENT LOADS

ABOVE GRADE WALLS

Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	N 166.8	Heating BTUH: Cooling BTUH:	987 326
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	E 135.3	Heating BTUH: Cooling BTUH:	800 265
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	W 105	Heating BTUH: Cooling BTUH:	621 205
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	N 159.6	Heating BTUH: Cooling BTUH:	944 312
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	E 80.1	Heating BTUH: Cooling BTUH:	474 157
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	W 77.7	Heating BTUH: Cooling BTUH:	460 152
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	E 84.7	Heating BTUH: Cooling BTUH:	501 166
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	S 220.5	Heating BTUH: Cooling BTUH:	1,304 431
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	W 87.7	Heating BTUH: Cooling BTUH:	519 172
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	W 127.9	Heating BTUH: Cooling BTUH:	757 250
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	E 155.1	Heating BTUH: Cooling BTUH:	917 303
Map trace wall Frame Wall, Wood framing, R-13 cavity insulation, Stucco or Siding.	Construction nr: U Value:	12C-0s w 0.091	Exposure: Area:	S 135.6	Heating BTUH: Cooling BTUH:	802 265

BELOW GRADE WALLS

There are no components for this section.

WINDOWS

Default small windows for wall id 9640634 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 N	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 80
--	--	--------------	-------------------	--------------	--------------------------------	-----------

WINDOWS						
Default medium windows for wall id 9640634 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 36 N	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	749 479
Default small windows for wall id 9640635 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 183
Default small windows for wall id 9640636 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 211
Default medium windows for wall id 9640636 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 24 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	499 843
Default small windows for wall id 9640612 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 N	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 80
Default small windows for wall id 9640613 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 183
Default medium windows for wall id 9640613 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 366
Default small windows for wall id 9640611 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 422
Default medium windows for wall id 9640611 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 422
Default small windows for wall id 9640656 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 366
Default medium windows for wall id 9640656 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 366
Default small windows for wall id 9640655 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 422
Default medium windows for wall id 9640655 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 12 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	250 422
Default small windows for wall id 9640658 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 211
Default medium windows for wall id 9640658 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 24 W	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	499 843
Default small windows for wall id 9640643 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 E	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 183
Default small windows for wall id 9640644 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 6 S	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	125 173
Default medium windows for wall id 9640644 Window, NFRC rated, Clear glass.	Construction nr: Area: Exposure:	1G 24 S	U Value: SHGC:	0.32 0.68	Heating BTUH: Cooling BTUH:	499 706

Window cooling BTUHs shown here are daily average values. See AED graphs for details of fenestration loads during the day.

CEILINGS						
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	359.8	Heating BTUH: Cooling BTUH:	468 360

CEILINGS					
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	289.3	Heating BTUH: 376 Cooling BTUH: 289
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	208	Heating BTUH: 270 Cooling BTUH: 208
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	301.2	Heating BTUH: 392 Cooling BTUH: 301
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	434.8	Heating BTUH: 565 Cooling BTUH: 435
Map trace generated ceiling Ceiling under attic or attic knee wall, Asphalt shingles, Dark, R-50.	Construction nr: U Value:	16B-50 ad 0.02	Area:	340.1	Heating BTUH: 442 Cooling BTUH: 340

SKYLIGHTS					
There are no components for this section.					

Skylight cooling BTUHs shown here are daily average values. See AED graphs for details of fenestration loads during the day.

DOORS					
There are no components for this section.					

FLOORS					
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 359.8	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	2,802 0 1.108
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 289.3	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	1,080 0 1.108
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 208	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	2,924 0 1.108
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 301.2	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	3,529 0 1.108
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 434.8	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	1,260 0 1.108
Map trace generated floor Concrete slab on grade floor, R-10 slab insulation.	Construction nr: Area:	22C-10pm 340.1	Heating U Value:0 Cooling U Value:0	Heating BTUH: Cooling BTUH: F Value:	2,614 0 1.108

VENTILATION					
Outside air damper	Heating SER: Sensible SER: Latent LER:	N/A N/A N/A	CFM:	70	Heating BTUH: 4,905 Sensible BTUH: 1,132

HOT WATER PIPING

There are no components for this section.

DUCTS

There are no components for this section.

INFILTRATION

Leakage Category:	Average	NCFM Heating:	110	Heating BTUH:	7,721
		NCFM Cooling:	58	Sensible BTUH:	938
				Latent BTUH:	1,005

BLOWER MOTOR

There are no components for this section.

WINTER HUMIDIFICATION

There are no components for this section.

OCCUPANTS

Nr. Occupants:	4	Sensible BTUH:	920	Latent BTUH:	800
Nr. Occupants:	4	Sensible BTUH:	920	Latent BTUH:	800

APPLIANCES

Color television	Quantity:	1	Sensible BTUH:	683
			Latent BTUH:	0
Computer and monitor	Quantity:	1	Sensible BTUH:	538
			Latent BTUH:	0
Light Fixture - 100W	Quantity:	3	Sensible BTUH:	1,023
			Latent BTUH:	0
Range with hood and exhaust to outdoors	Quantity:	1	Sensible BTUH:	0
			Latent BTUH:	0
Dishwasher	Quantity:	1	Sensible BTUH:	1,024
			Latent BTUH:	358
Refrigerator or Freezer - 16CuFt	Quantity:	1	Sensible BTUH:	1,000
			Latent BTUH:	0
Color television	Quantity:	1	Sensible BTUH:	683
			Latent BTUH:	0
Computer and monitor	Quantity:	1	Sensible BTUH:	538
			Latent BTUH:	0
Light Fixture - 100W	Quantity:	3	Sensible BTUH:	1,023
			Latent BTUH:	0

PLANTS

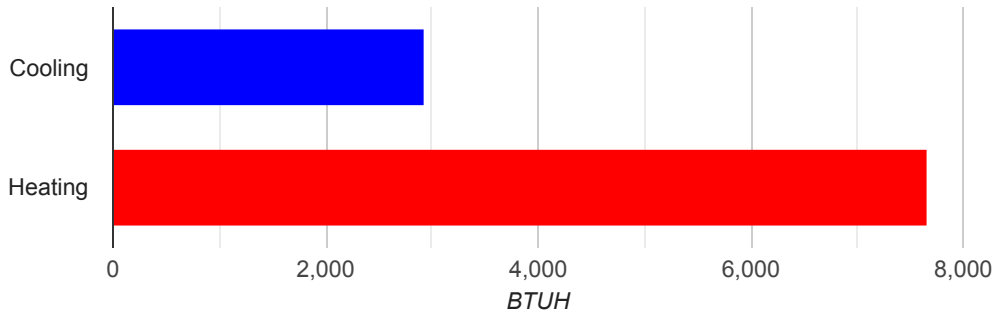
Plant Size:	medium	Quantity:	5	Latent BTUH:	100
Plant Size:	large	Quantity:	5	Latent BTUH:	150
Plant Size:	medium	Quantity:	5	Latent BTUH:	100
Plant Size:	large	Quantity:	5	Latent BTUH:	150

ROOM DETAIL

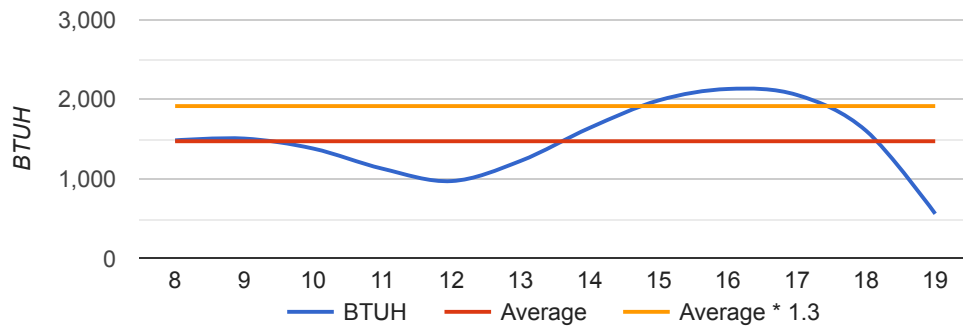
Room name: Entry

Heated square footage:	208	Total Cooling BTUH:	2,916
Cooled square footage:	208	Total Heating BTUH:	7,659
Heated volume (above grade CF):	1,872	CFM:	136
Cooled volume (above grade CF):	1,872		
Exposed wall area (SF):	365		

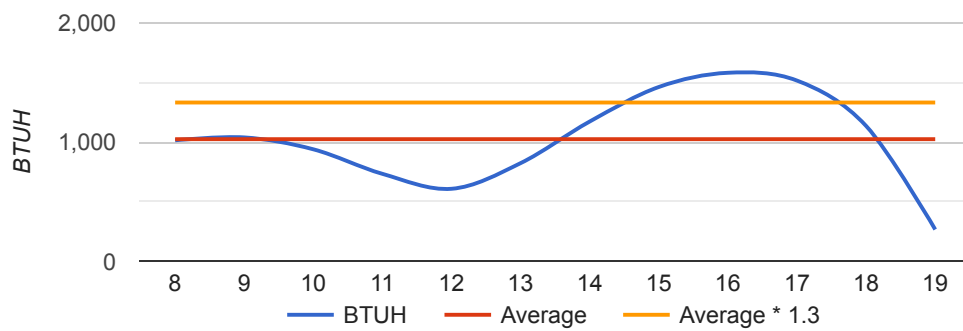
Load Calculation



AED Graph (mid-summer)



AED graph (fall)

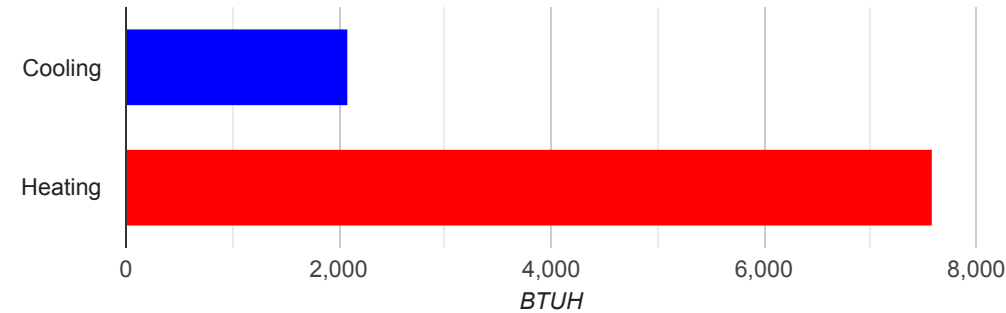


ROOM DETAIL

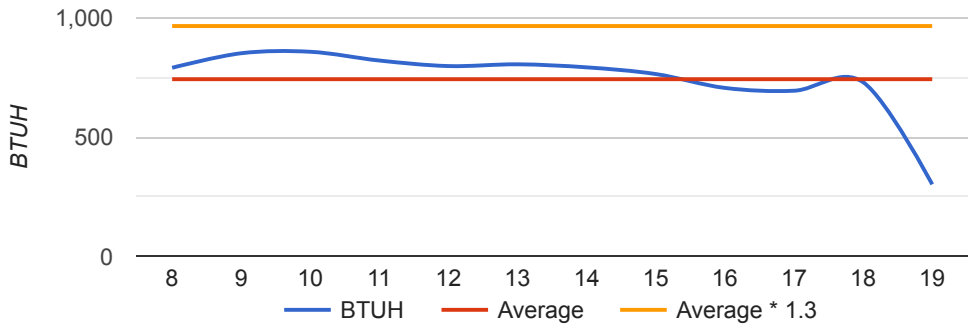
Room name: Master

Heated square footage:	360	Total Cooling BTUH:	2,076
Cooled square footage:	360	Total Heating BTUH:	7,577
Heated volume (above grade CF):	3,238	CFM:	94
Cooled volume (above grade CF):	3,238		
Exposed wall area (SF):	350		

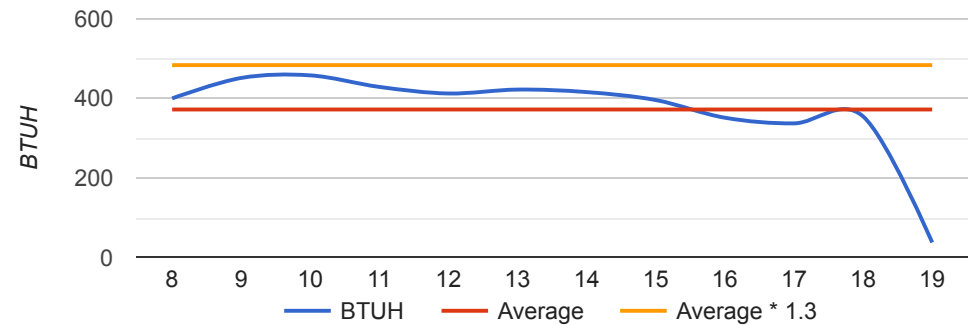
Load Calculation



AED Graph (mid-summer)



AED graph (fall)

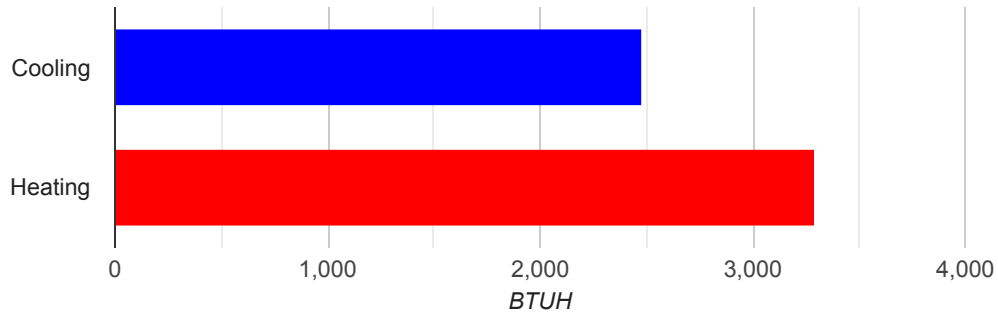


ROOM DETAIL

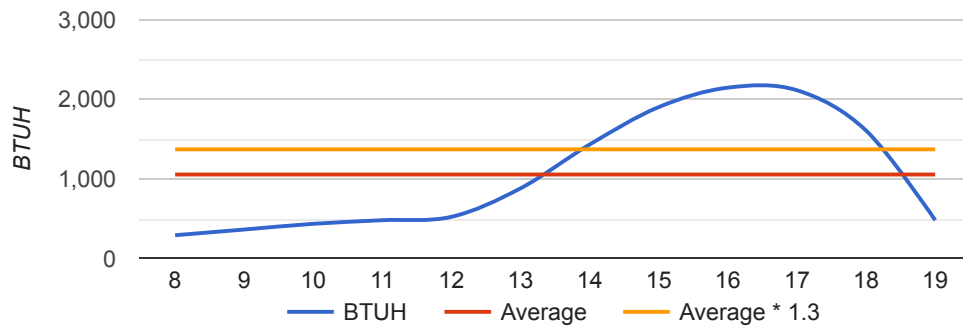
Room name: Bedroom-1

Heated square footage:	289	Total Cooling BTUH:	2,472
Cooled square footage:	289	Total Heating BTUH:	3,288
Heated volume (above grade CF):	2,604	CFM:	120
Cooled volume (above grade CF):	2,604		
Exposed wall area (SF):	135		

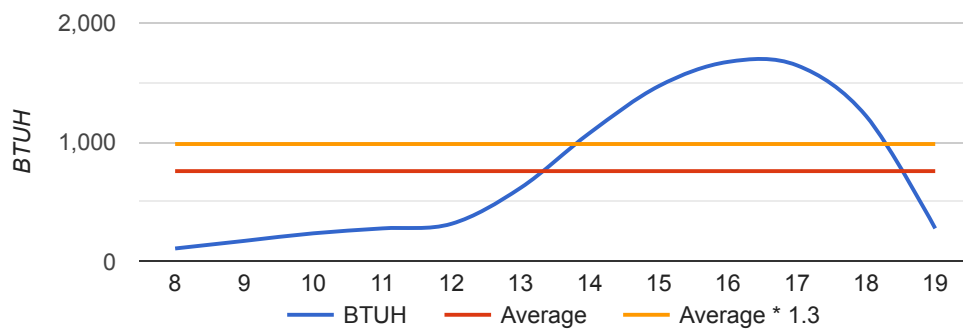
Load Calculation



AED Graph (mid-summer)



AED graph (fall)

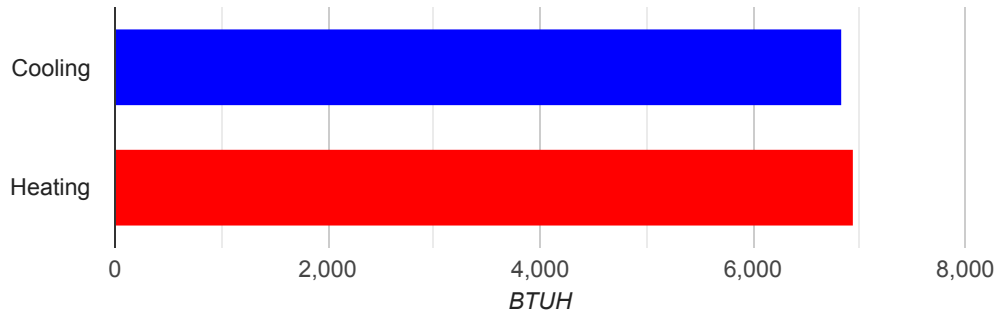


ROOM DETAIL

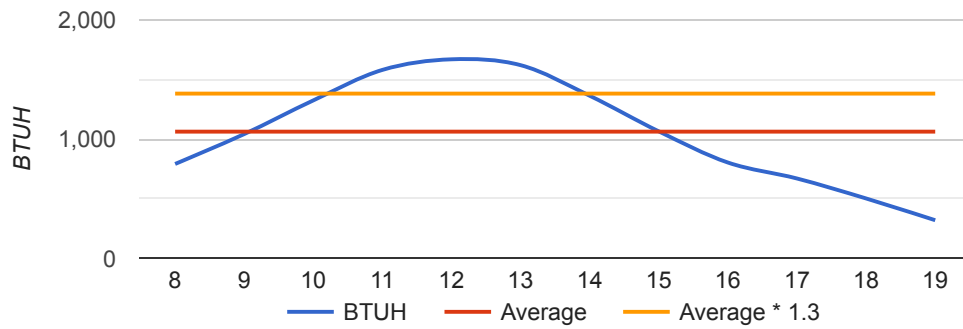
Room name: Living Room

Heated square footage:	340	Total Cooling BTUH:	6,829
Cooled square footage:	340	Total Heating BTUH:	6,945
Heated volume (above grade CF):	3,061	CFM:	254
Cooled volume (above grade CF):	3,061		
Exposed wall area (SF):	327		

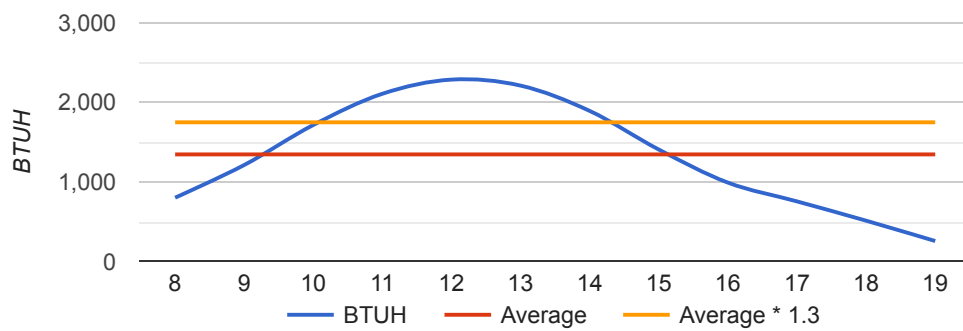
Load Calculation



AED Graph (mid-summer)



AED graph (fall)

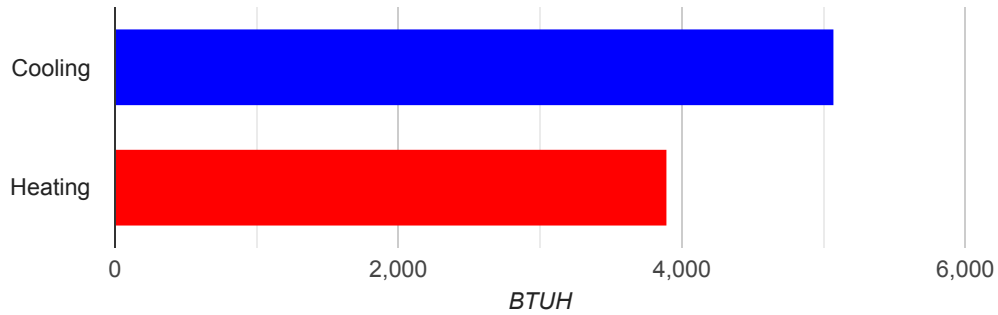


ROOM DETAIL

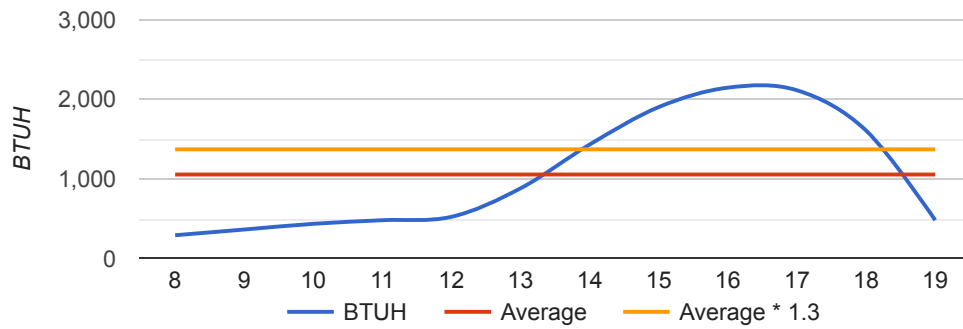
Room name: Kitchen

Heated square footage:	435	Total Cooling BTUH:	5,069
Cooled square footage:	435	Total Heating BTUH:	3,893
Heated volume (above grade CF):	3,913	CFM:	233
Cooled volume (above grade CF):	3,913		
Exposed wall area (SF):	158		

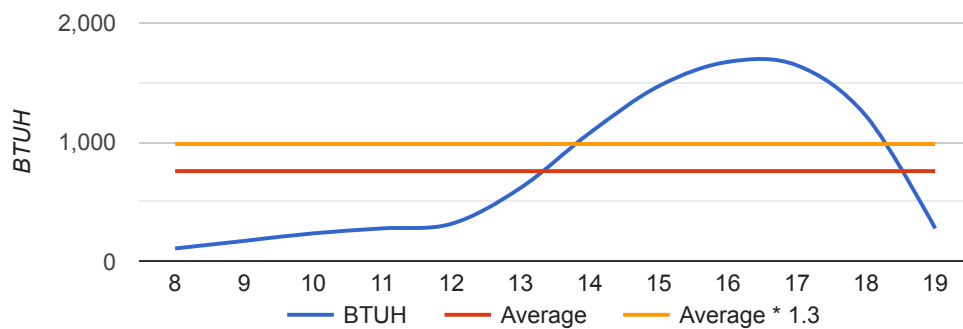
Load Calculation



AED Graph (mid-summer)



AED graph (fall)

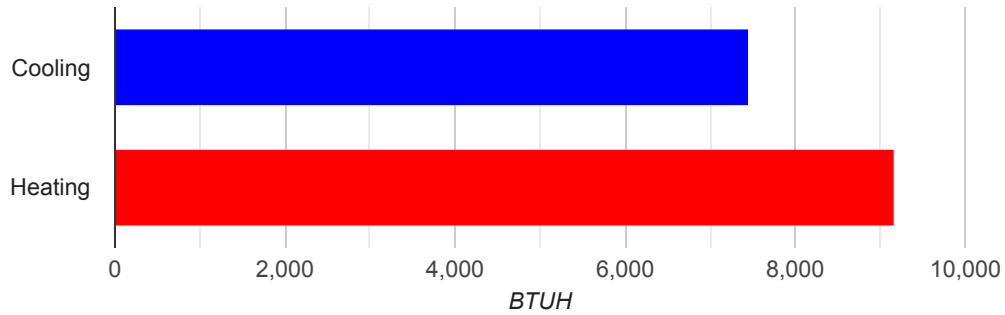


ROOM DETAIL

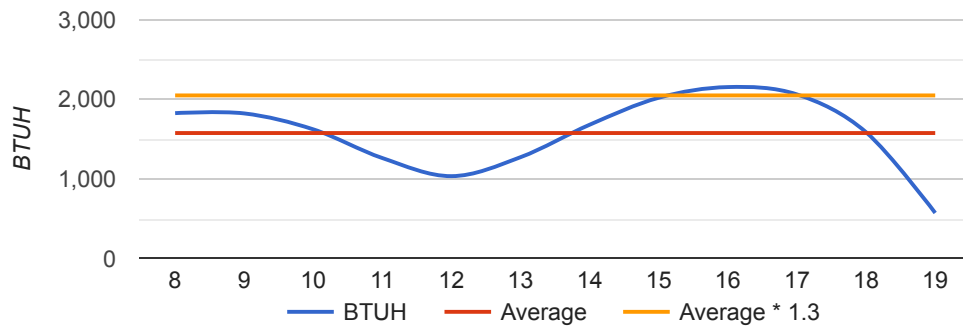
Room name: Family Room

Heated square footage:	301	Total Cooling BTUH:	7,446
Cooled square footage:	301	Total Heating BTUH:	9,160
Heated volume (above grade CF):	2,711	CFM:	279
Cooled volume (above grade CF):	2,711		
Exposed wall area (SF):	441		

Load Calculation



AED Graph (mid-summer)



AED graph (fall)

